GLA UNIVERSITY, GREATER NOIDA

Greater Noida – 201310, Uttar Pradesh, India



Title: Next-Song Recommender for a Music Streaming App

SOFTWARE REQUIREMENT SPECIFICATION

Version 1.0

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING - AI

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GLA UNIVERSITY, GREATER NOIDA – 201310

<MONTH, YEAR OF SUBMISSION>

Software Requirements Specification

Next-Song Recommender v1.0

Revision History

Version	Date	Change	Reason
1.0	**04-09-2026**	Initial version	First submission

1. Introduction

1.1 Purpose

This document specifies the requirements of the Next-Song Recommender v1.0, a supervised machine learning–based recommendation tool for music streaming. It serves as the baseline against which implementation and test cases will be verified.

1.2 Scope

The system recommends a ranked Top-K list of songs similar to a selected track using audio features (danceability, energy, tempo, valence, etc.). Users can interactively control recommendations through mood-based and metadata-based filters. The system does not predict actual user listening behavior since the dataset lacks user history.

1.3 Definitions

Term	Definition
Mood Shift	Scheduled change in emotional tone of recommendations after N songs.
Preference Selector	User option to filter next songs by tone, language, singer, or hero.
Top-K	Ranked list of K recommended songs.
SMOTE	Synthetic Minority Oversampling Technique for class balancing.

2. Overall Description

2.1 User Classes

User Class	Who They Are	What They May Do
Listener	End-user of the music app	Select songs, choose mood shift, set preference filters, view recommendations

2.2 Assumptions and Dependencies

- Depends on Spotify Tracks Dataset (**##dataset URL##**).
- Depends on Streamlit framework for UI.
- Relies on scikit-learn for ML models.

2.3 Constraints

- C1: Dataset does not contain user listening history.
- C2: Limited to supervised classification on audio features.

3. Data Source and Baseline

Item	Value
Dataset name	**##Spotify Tracks Dataset - confirm exact dataset name##**
Source and URL	**##Kaggle / other - insert exact URL##**
Licence / terms of use	**##Insert licence / terms of use##**
Number of rows and columns	**##Fill actual numbers, from opening the file##**
Prediction target	Song suitability (match to selected track)
Task type	Classification
Class balance	**##Fill percentages##**
Primary evaluation metric	F1-score
Why that metric	Balances precision and recall for imbalanced classes
Level 0 baseline and its score	**##insert score##**
Known data quality problems	Missing values, duplicates, outliers
Level 1 baseline and its score	**##Insert a simple model (e.g., logistic/linear regression or a shallow tree) and its score##**
Target for this release	**##Insert the score the model must beat to be worth shipping##**

4. System Architecture

##Insert Figure 1: System architecture diagram (draw.io / Excalidraw) here##

Component	Responsibility
Data Preprocessor	Cleans dataset, handles missing values
Feature Engineer	Extracts predictive audio features
Model Trainer	Trains supervised models (RF, SVM, GB)
Recommendation Engine	Generates Top-K ranked list
Mood Controller	Implements scheduled mood shift
Preference Selector	Filters songs by tone, language, singer, hero
Streamlit UI	Interactive interface for users

5. Functional Requirements

5.1 Data Upload and Validation

FR-1.1 The system shall accept a valid dataset file and reject invalid formats with an error message.

FR-1.2 The system shall clean missing values and duplicates.

FR-1.3 The system shall normalize audio features before training.

FR-1.4 The system shall reject datasets missing mandatory columns and display the missing column name.

5.2 Model-backed Requirements

FR-2.1 The system shall recommend Top-K songs based on audio similarity.

FR-2.2 The system shall allow users to select mood shift intervals (10 or 20 songs).

FR-2.3 The system shall allow users to select preference filters (tone, language, singer, hero).

FR-2.4 The system shall provide fallback recommendations using popularity baseline if the ML model fails.

FR-2.5 The system shall display confidence scores with recommendations.

FR-2.6 The system shall log each recommendation session with user choices.

FR-2.7 The system shall allow the Listener to **##override or manually adjust the recommended list (e.g., remove/replace a song, re-rank results)##** — added because human override of model output was missing from the original list and is commonly required at review.

6. Non-Functional Requirements

NFR-1 The system shall return recommendations within 3 seconds for datasets up to 50,000 tracks.

NFR-2 The system shall use GitHub for version control.

NFR-3 The system shall log recommendation requests with unique IDs.

NFR-4 The system shall maintain 95% uptime during demo sessions.

NFR-5 The system shall support at least 100 concurrent users during demo testing.

NFR-6 The system shall **##name a specific security mechanism, e.g., HTTPS in transit, hashed/salted credentials, input sanitisation on uploads##** — added because a named security requirement was missing from the original list.

7. Build Plan

7.1 MVP

FR/NFR ID	Requirement	Why in MVP
FR-1.1	Dataset upload validation	Without it, downstream fails
FR-1.2	Data cleaning	Ensures usable dataset
FR-2.1	Top-K recommendation	Core functionality
FR-2.4	Fallback baseline	Guarantees usable output

FR/NFR ID	Requirement	Why in MVP
FR-2.2	Mood shift option	Differentiator feature
NFR-1	Response time	Demonstrable performance
NFR-2	GitHub version control	Ensures collaboration

Deliberately not in the MVP: ##Name one requirement that felt important but did not make the cut, and what you would need to know before adding it##

7.2 Build Order

Fixed project stages (given):

Stage	Deliverable	Date
Development Sprint I	Prototype / initial working model	27 Sep 2026
Development Sprint II	50% functional application	17 Oct 2026
Development Sprint III	75–80% functional application	7 Nov 2026
Testing & Validation	Test report and bug log	17 Nov 2026
Deployment & Documentation	Deployed application, user manual, technical documentation	30 Nov 2026
Final Presentation	Final report, presentation and viva	5–10 Dec 2026

Your plan (every FR/NFR ID from Section 5 and 6 should appear in exactly one row):

Stage / Date	What Will Be Working	FR/NFR IDs	How I Will Know It's Done
Sprint I — 27 Sep	MVP: dataset upload, validation, cleaning, Top-K recommendation, popularity fallback, mood shift option	FR-1.1, FR-1.2, FR-2.1, FR-2.4, FR-2.2, NFR-1, NFR-2	**##What you would demonstrate##**
Sprint II — 17 Oct	**##What is added##**	**##IDs##**	**##What you would demonstrate##**
Sprint III — 7 Nov	**##What is added##**	**##IDs##**	**##What you would demonstrate##**
Testing — 17 Nov	**##What is tested, not what is built##**	**##IDs##**	**##What you would demonstrate##**
Deployment — 30 Nov	**##Live URL and documentation##**	**##IDs##**	**##What you would demonstrate##**

8. Data Requirements

8.1 Entities

Entity: Track

Field	Type	Constraint
id	UUID	Primary key

Field	Type	Constraint
title	String	Not null
artist	String	Not null
language	String	Optional
audio_features	JSON	Must include valence, energy, tempo

Other entities: User, RecommendationSession

8.2 Entity Relationship Diagram

Not required for v1.0. **##Insert an ER diagram covering every entity in 8.1 (Track, User, RecommendationSession) with cardinalities marked on every relationship, in a later version##**

9. Design Decisions and Alternatives Considered

#	Decision	Alternative Rejected	Reason
D1	Chose supervised classification	Unsupervised clustering	Dataset has labeled subsets
D2	Chose Streamlit	Flask	Faster prototyping for interactive UI
D3	Added Mood Shift feature	Static playlists	Unique differentiator vs YouTube Music

10. Acceptance Criteria

AC-1 Upload valid dataset → system accepts and preprocesses.

AC-2 Select a song → system returns Top-K recommendations.

AC-3 Model unavailable → system falls back to popularity baseline.

AC-4 After 10 songs → system prompts for mood shift.

AC-5 User selects "same singer" → recommendations filtered accordingly.

11. Out of Scope for This Release

Predicting actual user listening behavior. Requires user history, which this dataset does not contain.

Real-time collaborative filtering. Deferred to a later release.

Mobile app deployment. Web-only for now.